

CSE400 – Milestone 4 Scribe (Q4–Q6)

S1_G7_ITS

Scribe Question 4: Implementation within the Simulation Pipeline

Overall Simulation Pipeline

The system follows a structured pipeline consisting of:

- **SUMO (Simulation of Urban Mobility):** Simulates microscopic traffic dynamics such as vehicle movement, queue formation, and speed variations.
- **Python Controller:** Implements the traffic signal decision logic.
- **TraCI Interface:** Enables real-time communication between SUMO and Python, allowing exchange of traffic state and control signals.

Integration of Randomized Algorithm

The randomized approach is incorporated within the **decision module**, which operates at each discrete time-step.

- Baseline: Deterministic rule (select lane with maximum queue).
- Modified: Replaced with a **hybrid Las Vegas algorithm**.

Changes in Code and Logic

- Introduced a **trial-based randomized selection process**:
 1. Identify top candidate lanes (high queue lengths)
 2. Perform multiple randomized trials
 3. Evaluate resulting queue states
 4. Select the best feasible decision
- Added a **validation mechanism**:
 - If a random decision is suboptimal → fallback to deterministic rule

Generation of Randomness

- **Vehicle arrivals:**

$$A_i \sim \text{Poisson}(\lambda)$$

- **Service rate:**

– Modeled using exponential or fixed distributions

- **Signal selection (X_i):**

– Randomly chosen among candidate lanes

Unchanged Components

- SUMO simulation environment
- TraCI communication interface
- Data collection and evaluation metrics

Conclusion

Only the decision logic is modified to include controlled randomness, while the rest of the pipeline remains unchanged. This ensures stability while improving adaptability.

Scribe Question 5: Evaluation Metrics and Comparative Results

Evaluation Metrics Used

- **Waiting Time (per vehicle):** Measures delay experienced before crossing the intersection.
- **Queue Length:** Number of vehicles waiting at a signal, indicating congestion.
- **Convergence Behavior:** Reflects how quickly the system stabilizes over time.
- **Speed Distribution:** Indicates smoothness and efficiency of traffic flow.

Why These Metrics Are Appropriate

- Waiting time directly captures user-level delay.
- Queue length reflects congestion at intersections.
- Convergence ensures stability of the system.
- Speed distribution measures flow efficiency.

Comparative Results

- Deterministic Approach: **2.05**
- Las Vegas Algorithm: **1.04**
- Improvement: **49.27%**

Interpretation of Results

- Nearly **50% reduction in waiting time.**
- Reduced queue lengths indicate lower congestion.
- Faster convergence shows improved system stability.
- Better speed distribution reflects smoother traffic flow.

Conclusion

The randomized (Las Vegas) approach significantly outperforms the deterministic baseline by adapting better to stochastic traffic conditions.

Scribe Question 6: Insights and Remaining Challenges

Key Insights

- Traffic systems are inherently **stochastic**, making deterministic control limited.
- Randomization improves **adaptability** under dynamic conditions.
- Prevents **lane starvation** by avoiding strict priority-based decisions.
- Hybrid approach (random + deterministic fallback) ensures both **performance and stability**.
- Leads to improved convergence and consistent long-term behavior.

When Randomized Strategy Performs Better

- High variability in vehicle arrivals
- Uneven traffic distribution across lanes
- Situations where deterministic rules become repetitive

Limitations

- Less effective in highly predictable traffic conditions
- Excess randomness can lead to unstable decisions

Remaining Challenges

- **Stability vs Randomness Tradeoff**
 - Too much randomness → instability
 - Too little → reduced benefit
- **Real-Time Constraints**
 - Decisions must be computed efficiently at each time-step
- **Reproducibility**
 - Results vary across runs, requiring averaging
- **Scalability**
 - Multi-intersection systems increase complexity
- **System Integration Issues**
 - TraCI communication delays and synchronization challenges

Final Insight

Controlled randomization using the hybrid Las Vegas approach balances exploration and stability, leading to improved traffic signal performance.